Geometry of Light on the Round Three-Sphere: Hopf–Jacobi Dynamics, Maxwell Intensity, Dual Tomography, and Nonlinear Spectral Certificates

Research working draft for the Batenin–Preece programme
Mathematical manuscript; authorship and independent review pending

October 10, 2026

Abstract

We develop an algebraically parameterized geometry-of-light research framework on the fixed round three-sphere. Smooth null Beltrami fields built from homogeneous binary forms provide a common mathematical input for two *distinct* dynamical equations: source-free Maxwell evolution and exact pressure-balanced viscous flow. For the latter, we establish both an unforced multispectral toroidal Jacobi family and a symmetry-protected nonresonant Hopf–Killing family, distinguishing spectral degeneracy from vanishing projected nonlinear interaction. We derive energy, helicity, enstrophy, pressure-beat and Noether identities, a signed-curl spectral variance, and physical-time Fubini–Study/Fisher formulas for reducing viscous histories. A positive moment–Hankel hierarchy gives finite-rank spectral certificates with stated conditioning limits. In an externally controlled fixed-Hopf sector, projective forcing has a Berry-geometric cost and permits a smooth exchange of linked nodal fibres. Our new cross-domain reconstruction theorem shows that $(m+1)(2m+1)$ calibrated equal-time intensity or pressure-deficit samples recover the degree-m binary form up to global phase and therefore recover its Hopf nodal divisor. A separate finite energy–helicity Prony procedure resolves occupied absolute curl shells and both chirality weights. An explicit counterexample proves that integrated energy–helicity histories alone cannot identify nodal topology, even without noise. The manuscript includes reproducible algebraic verification and sharply typed transfer gates from OpenAI Math. Polynomial optical knots, stellar representations and moment inversion have substantial prior art; the worldwide novelty of the joint construction remains unestablished. We make no claim of Maxwell–Navier–Stokes physical identity, universal PDE regularity, quantum-gravity closure or observed cosmological topology.

1 Scope, conventions, and motivation

An exact solution of a nonlinear PDE in a selected invariant family is not a regularity theorem for arbitrary initial data. Our aim is instead a reproducible atlas in which spectral, pressure, dynamical, topological, and projective data are connected by explicit mathematical arrows. The initial motivation is the fixed-Hopf null curl-eigenfield family of Batenin and Preece [4], a recent unreviewed preprint. Earlier work on Beltrami mechanics, Majorana stars and Berry geometry must be separated from any particular hydrodynamic packaging [5, 6, 7]. Contemporary work on smooth forcing in flat Navier–Stokes systems has a different setting and goal [10].

Fix the unit-round or radius-R sphere

$$S^3_R = \{(z_1, z_2) \in \mathbb{C}^2 : \ |z_1|^2 + |z_2|^2 = 1\}, \quad g_R = R^2 g_1. \tag{1}$$

The z coordinates are dimensionless. We use the positive Hodge Laplacian $\Delta_H = \mathrm{d}\delta + \delta\mathrm{d}$ on one-forms, transferred to vector fields by musical isomorphisms. For divergence-free velocity u,

$$\partial_t u + \nabla_u u = -\operatorname{grad} P - \nu \Delta_H u + f, \qquad \operatorname{div} u = 0, \quad \nu > 0. \tag{2}$$

On divergence-free fields $\Delta_H u = \operatorname{curl}^2 u$. This is a specified curved-space viscosity convention, not a claim that all definitions of the viscous stress tensor on curved manifolds are identical.

2 An unforced multispectral toroidal sector

2.1 Jacobi eigenbasis on S^3_R

Away from the two collapsed coordinate circles, write

$$z_1 = \sqrt{1-x}\, e^{i\phi}, \qquad z_2 = \sqrt{x}\, e^{i\psi}, \quad 0 < x < 1, \tag{3}$$

so that

$$g_R = R^2 \left[\frac{\mathrm{d}x^2}{4x(1-x)} + (1-x)\mathrm{d}\phi^2 + x\mathrm{d}\psi^2 \right], \qquad \mathrm{d}V = \frac{R^3}{2}\mathrm{d}x\mathrm{d}\phi\mathrm{d}\psi. \tag{4}$$

Let $A(x), B(x)$ be smooth polynomials. The field

$$u = A(x)\partial_\phi + B(x)\partial_\psi \tag{5}$$

is globally smooth, tangent to the Clifford tori $x = \text{const}$, and divergence-free. Its integral curves have constant x; the angular speeds are A and B. We fix the outward-boundary orientation induced by the standard real coordinates $(\Re z_1, \Im z_1, \Re z_2, \Im z_2)$. In these Hopf coordinates its positive volume form is $-\mathrm{d}x \wedge \mathrm{d}\phi \wedge \mathrm{d}\psi$, up to the positive scale $R^3/2$. This orientation is consistent with the positive-curl quaternion frame $X_1 = iq$, $X_2 = jq$, $X_3 = kq$ used below. The unsigned volume element remains $(R^3/2)\mathrm{d}x\mathrm{d}\phi\mathrm{d}\psi$.

Define $J^A_n = P^{(0,1)}_n(1-2x)$ and $J^B_n = P^{(1,0)}_n(1-2x)$ using Jacobi polynomials, with $J^A_0 = J^B_0 = 1$. Set

$$U^A_n = J^A_n(x)\partial_\phi, \qquad U^B_n = J^B_n(x)\partial_\psi, \qquad \kappa_n = \frac{2(n+1)}{R}. \tag{6}$$

Lemma 2.1 (Curl and Hodge eigenidentities)**.** *For all $n \geq 0$,*

$$\operatorname{curl} U^A_n = +\kappa_n U^B_n, \qquad \operatorname{curl} U^B_n = +\kappa_n U^A_n, \quad \Delta_H U^{A,B}_n = \kappa_n^2 U^{A,B}_n. \tag{7}$$

The fields $U^A_n \pm U^B_n$ have curl eigenvalue $\pm\kappa_n$.

Proof. If $\alpha = u^\flat = R^2[(1-x)A\mathrm{d}\phi + xB\mathrm{d}\psi]$, direct application of $\star\mathrm{d}$ gives

$$\operatorname{curl} u = +\frac{2}{R}(xB)'\partial_\phi - \frac{2}{R}((1-x)A)'\partial_\psi. \tag{8}$$

The Jacobi derivative identities

$$((1-x)J^A_n)' = -(n+1)J^B_n, \qquad (xJ^B_n)' = (n+1)J^A_n \tag{9}$$

yield the first claims. Applying curl twice gives the last claim by the divergence-free Hodge identity. □

Theorem 2.2 (Unforced multi-rate Jacobi solution)**.** *For any integer $N \geq 0$ and arbitrary real coefficients a_n, b_n, define*

$$A(t,x) = \sum_{n=0}^{N} a_n e^{-\gamma_n t} J^A_n(x), \qquad B(t,x) = \sum_{n=0}^{N} b_n e^{-\gamma_n t} J^B_n(x), \tag{10}$$

$$\gamma_n = \nu\kappa_n^2 = \frac{4\nu(n+1)^2}{R^2}.$$

Then for all $t \geq 0$ the globally smooth velocity $u(t) = A\partial_\phi + B\partial_\psi$ and pressure

$$\boxed{P(t,x) = C(t) - \frac{R^2}{2}\int_0^x [A(t,y)^2 - B(t,y)^2]\mathrm{d}y} \tag{11}$$

solve Eq. (2) with exactly zero force $f = 0$. Arbitrarily many distinct nonzero eigenvalues κ_n^2 may be present.

Proof. The velocity is invariant under both angular circle actions; it is divergence-free by the invariant volume density in Eq. (4). Its convective acceleration is purely in the x direction. In the angle χ defined by $x = \sin^2\chi$, the only relevant Christoffel symbols are $\Gamma^\chi_{\phi\phi} = \sin\chi\cos\chi$ and $\Gamma^\chi_{\psi\psi} = -\sin\chi\cos\chi$. Hence

$$\nabla_u u = \sin\chi\cos\chi(A^2 - B^2)\partial_\chi, \qquad \partial_x P = -\frac{R^2}{2}(A^2 - B^2), \tag{12}$$

which implies $\nabla_u u = -\operatorname{grad} P$. Lemma 2.1 and the exponentials give $\partial_t u = -\nu\Delta_H u$; adding these two identities proves the full nonlinear equation. All fields are smooth at $x = 0, 1$ because $x = |z_2|^2$ and $\partial_\phi, \partial_\psi$ extend as globally smooth Killing fields, vanishing appropriately at the collapsed circles. □

2.2 Integral characteristics and pressure beat spectroscopy

The Jacobi norm identities give

$$\int_0^1 (1-x)J_n^A J_k^A \mathrm{d}x = \int_0^1 x J_n^B J_k^B \mathrm{d}x = \frac{\delta_{nk}}{2(n+1)}. \tag{13}$$

Corollary 2.3 (Energy, helicity, enstrophy, and Killing projections)**.** *Writing $\mathcal{E} = \frac{1}{2}\int |u|^2 \mathrm{d}V$, $\mathcal{H} = \int u \cdot \operatorname{curl} u \mathrm{d}V$, $\mathcal{Z} = \int |\operatorname{curl} u|^2 \mathrm{d}V$, and $Q_K = \int g_R(u, K)\mathrm{d}V$, the solution of Theorem 2.2 has*

$$\mathcal{E}(t) = \frac{\pi^2 R^5}{2}\sum_{n=0}^{N} \frac{a_n^2 + b_n^2}{n+1} e^{-2\gamma_n t}, \tag{14}$$

$$\mathcal{H}(t) = +4\pi^2 R^4 \sum_{n=0}^{N} a_n b_n e^{-2\gamma_n t}, \tag{15}$$

$$\mathcal{Z}(t) = 4\pi^2 R^3 \sum_{n=0}^{N} (n+1)(a_n^2 + b_n^2) e^{-2\gamma_n t}, \tag{16}$$

$$Q_{\partial_\phi} = \pi^2 R^5 a_0 e^{-\gamma_0 t}, \qquad Q_{\partial_\psi} = \pi^2 R^5 b_0 e^{-\gamma_0 t}. \tag{17}$$

In particular $\mathrm{d}\mathcal{E}/\mathrm{d}t = -\nu\mathcal{Z}$ and $\mathrm{d}Q_K/\mathrm{d}t = -4\nu Q_K/R^2$ for the displayed Killing fields. The individual mode helicity polarizations $+2a_n b_n/(a_n^2 + b_n^2)$ are invariant whenever their denominators are nonzero.

Proof. Insert the volume element and the pointwise norm $|u|^2 = R^2[(1-x)A^2 + xB^2]$ into the definitions and apply Eq. (13). Curl acts by the off-diagonal matrix of Lemma 2.1, immediately yielding Eq. (15). The Killing projections select only $n = 0$ by Jacobi orthogonality against constants. Differentiating Eq. (14) gives the energy identity using $\gamma_n = \nu\kappa_n^2$. □

The pressure is not an independent spectral mode. Expanding Eq. (11) shows mixed exponents $\exp[-(\gamma_n + \gamma_k)t]$, with cross terms even when $\kappa_n \neq \kappa_k$:

$$-\frac{2}{R^2}\partial_x P = \sum_{n,k=0}^{N} e^{-(\gamma_n+\gamma_k)t}\big[a_n a_k J_n^A J_k^A - b_n b_k J_n^B J_k^B\big]. \tag{18}$$

The pressure therefore registers the nonlinear cross-mode geometry while the velocity experiences *no* nonlinear spectral transfer. Equal sums $\gamma_n + \gamma_k$ cause possible spectral aliasing, and some cross coefficients can cancel. Consequently pressure measurements are not claimed to uniquely reconstruct all a_n, b_n in this sector.

Orientation audit. The signed-curl formulas and the quaternion frame must use the same outward-boundary orientation. Reversing the orientation in only one sector changes the signed eigenvalue of that sector and invalidates the stated hybrid resonance. With the orientation fixed above, the positive toroidal branch is the sum of the two Jacobi generators rather than their difference. The unsigned energy and enstrophy formulas are unchanged, whereas the signed helicity and its reconstruction formula reverse sign relative to the opposite convention.

3 A distinct pressure-observable Hopf sector

We now turn to a different exact family. It meets the toroidal one in simple Hopf/Killing cases but should not be confused with the entire Jacobi family. Fix a global oriented orthonormal Hopf frame (X_1, X_2, X_3) on S^3_R, with the common outward-boundary positive curl convention fixed in Sec. 2, and a homogeneous polynomial

$$p(z_1, z_2) = \sum_{k=0}^{m} c_k z_1^{m-k} z_2^k, \qquad b_k = \frac{c_k}{\sqrt{\binom{m}{k}}}, \qquad r = \|b\|. \tag{19}$$

For $m \geq 1$ define $\mathscr{U}[p] = \mathrm{Re}[p(X_2 + iX_3)]$. The selected Hopf identities are $\operatorname{div} \mathscr{U}[p] = 0$ and $\operatorname{curl} \mathscr{U}[p] = \lambda_m \mathscr{U}[p]$, $\lambda_m = (m+2)/R$. They can be checked directly in the global frame for these polynomials.

Explicit frame and atlas projection. For reproducibility set $R = 1$ and identify a unit quaternion with the ambient vector $q = (a, b, c, d)$, where $z_1 = a + ib$, $z_2 = c + id$. The oriented orthonormal quaternion frame is

$$X_1 = (-b, a, -d, c), \quad X_2 = (-c, d, a, -b), \quad X_3 = (-d, -c, b, a). \tag{20}$$

Consequently $\mathscr{U}[p] = (\Re p)X_2 - (\Im p)X_3$ and the toroidal components are $\partial_\phi = (-b, a, 0, 0)$, $\partial_\psi = (0, 0, -d, c)$. All are tangent to S^3. The atlas integrates the resulting tangent vector field on S^3 and displays the spatial stereographic projection

$$\Pi(q) = \frac{(b, c, d)}{1 - a}, \qquad q \neq (1, 0, 0, 0), \tag{21}$$

with only a fixed visual scale factor. The projection is exact away from its pole, while individual plotted integral curves are approximated by normalized fourth-order Runge–Kutta steps. In particular a graphically observed crossing of projected traces is not asserted to prove a linking invariant. The kinetic metric satisfies

$$\int_{S^3_R} |\mathscr{U}[p]|^2 \mathrm{d}V = C_m \|b\|^2, \qquad C_m = \frac{2\pi^2 R^3}{m+1}. \tag{22}$$

Lemma 3.1 (Realization of coefficient dynamics by solenoidal forcing)**.** *For an arbitrary C^1 coefficient path $b(t) \neq 0$ at a fixed m, the fields*

$$u(t) = \mathscr{U}[p(t)], \quad P(t) = -\frac{1}{2}|u(t)|^2, \quad f(t) = \mathscr{U}[\dot{p}(t) + \gamma p(t)], \quad \gamma = \nu\lambda_m^2 \tag{23}$$

solve Eq. (2) exactly, with $\operatorname{div} f = 0$. *The pressure-deficit profile* $2(P_{\max} - P) = |p(t,z)|^2$ *is constant along Hopf fibres and determines* $b(t)b(t)^\dagger$ *when* m *is known, as established for the preceding pressure-tomography family.*

Proof. A real Beltrami eigenfield obeys $\nabla_u u = \operatorname{grad}(|u|^2/2)$, with all curvature effects included in the Riemannian cross-product identity. Thus the pressure cancels nonlinear advection. Since $\Delta_H u = \lambda_m^2 u$, the chosen force is exactly the remaining term $\partial_t u + \nu\lambda_m^2 u$. Every coefficient derivative remains in the solenoidal curl-eigenspace. For $m \geq 1$ the polynomial has a projective root and hence $P_{\max} = 0$ in this pressure gauge. The fixed-degree polynomial intensity determines the rank-one coefficient projector by polarization of the homogeneous Hermitian polynomial; see also the numerical reconstruction in the companion research package. □

3.1 Projective metric, Berry connection, and pressure-only control bound

Define the normalized rank-one matrix $\rho = bb^\dagger/r^2$. It is determined by the normalized pressure deficit, although the complex global phase of b is not. Its Fubini–Study speed is

$$v_{\mathrm{FS}}^2 = \frac{1}{2}\operatorname{Tr}[(\dot\rho)^2] = \|\dot\zeta\|^2 - |\zeta^\dagger\dot\zeta|^2, \qquad \rho = \zeta\zeta^\dagger, \quad \|\zeta\| = 1. \tag{24}$$

The connection on unit coefficient lifts is $a(t) = \operatorname{Im}(\zeta^\dagger\dot\zeta)$. The quantity $\beta = \int_0^T a(t)\mathrm{d}t \bmod 2\pi$ is gauge invariant for a smooth periodic lift $\zeta(0) = \zeta(T)$; its sign depends on the Berry connection convention, but the distance to $2\pi\mathbb{Z}$ does not.

Theorem 3.2 (Exact pressure-observable actuation decomposition)**.** *For any nonzero* C^1 *coefficient path, write* $b(t) = r(t)e^{i\theta(t)}\zeta(t)$, *with* $\|\zeta\| = 1$. *The squared* L^2 *norm of the exact force in Lemma 3.1 obeys the identity*

$$\boxed{\frac{\|f(t)\|^2_{L^2(S^3_R)}}{C_m} = (\dot r + \gamma r)^2 + r^2 v_{\mathrm{FS}}^2 + r^2(\dot\theta + a)^2.} \tag{25}$$

Suppose additionally that $r(t) = r > 0$ *is constant and the pressure projector* $\rho(t)$ *is periodic with period* T, *while the full physical velocity also closes,* $u(T) = u(0)$. *Then*

$$\boxed{\int_0^T \|f(t)\|^2_{L^2}\mathrm{d}t \;\geq C_m r^2\left[\gamma^2 T + \frac{L_{\mathrm{FS}}^2 + \operatorname{dist}(\beta, 2\pi\mathbb{Z})^2}{T}\right],} \tag{26}$$

where $L_{\mathrm{FS}} = \int_0^T v_{\mathrm{FS}}(t)\mathrm{d}t$. *For a prescribed constant-speed projective loop the bound is achieved by an appropriate phase lift chosen so that the vertical connection velocity is constant and of minimum possible absolute winding. More generally the exact lower bound for a prescribed time-parametrized loop replaces* L_{FS}^2/T *by* $\int_0^T v_{\mathrm{FS}}^2\mathrm{d}t$.

Proof. The isometry Eq. (22) applied to the force gives $\|f\|^2 = C_m\|\dot b + \gamma b\|^2$. Decompose the coefficient derivative into the radial part, the component along $i\zeta$, and the complex-horizontal component. The three real directions are orthogonal, giving Eq. (25). For a periodic projective curve choose a periodic normalized representative $\zeta(t)$ on the time circle, which exists because every complex line bundle over S^1 is trivial. Velocity closure implies $e^{i\theta(T)} = e^{i\theta(0)}$, so $\int_0^T \dot\theta\mathrm{d}t = 2\pi n$. Therefore

$$\int_0^T (\dot\theta + a)^2\mathrm{d}t \geq \frac{(2\pi n + \beta)^2}{T} \geq \frac{\operatorname{dist}(\beta, 2\pi\mathbb{Z})^2}{T}.$$

A separate Cauchy–Schwarz inequality gives $\int_0^T v_{\mathrm{FS}}^2\mathrm{d}t \geq L_{\mathrm{FS}}^2/T$. Equality requires constant projective speed and constant vertical connection velocity, both achieved by time reparametrization and a smooth phase adjustment with the minimizing integer winding. The physical phase cannot simply be quotiented away in Eq. (2); its rotation changes u although not the pressure. This is the reason the Berry term carries a real forcing cost. □

3.2 An exactly saturating two-link braid

For $m = 2$, let

$$\boxed{p_t(z_1, z_2) = \frac{z_1^2 - e^{2\pi i t/T} z_2^2}{\sqrt{2}}, \qquad 0 \le t \le T.} \tag{27}$$

Its projective roots are $z_1/z_2 = \pm e^{\pi i t/T}$. Both roots remain distinct and exchange over a closed cycle; their inverse Hopf fibres remain linked with linking number $+1$ under the standard orientation. The coefficient lift $b = (1, 0, -e^{2\pi i t/T})/\sqrt{2}$ is physically periodic and normalized. Direct evaluation gives

$$L_{\mathrm{FS}} = \pi, \quad \beta = \pi \pmod{2\pi}, \quad \int_0^T \|f\|_{L^2}^2 \mathrm{d}t = \boxed{C_2 \left(\gamma^2 T + \frac{2\pi^2}{T}\right)}, \qquad \gamma = \frac{16\nu}{R^2}. \tag{28}$$

Thus this exact Navier–Stokes protocol saturates the Berry–Fubini–Study forcing lower bound. It is a forced protocol, not a claim that an unforced viscous flow can braid its pressure zeros. The braid is a loop in the configuration space of distinct unordered roots; all pairwise spatial Hopf linking numbers remain unchanged.

4 Curvature viscosity bridge and Noether charges

The operator in Eq. (2) is the Hodge Laplacian, whereas the Ebin–Marsden deformation-tensor dissipation considered in other work on round S^3 is a different physical choice. They must not silently be conflated [8, 9]. We fix the convention

$$A_{\mathrm{EM}} = -2 \operatorname{div} \operatorname{Def} = \Delta_H - 2 \operatorname{Ric} = \Delta_H - \frac{4}{R^2} I \tag{29}$$

on divergence-free vector fields. Here $\operatorname{Def} u$ is the symmetric covariant velocity gradient. Since $\operatorname{Ric} = 2g_R/R^2$, the curvature shift is scalar on the entire round sphere.

Theorem 4.1 (Exact pressure-balanced viscosity conjugacy)**.** *Assume a smooth solenoidal field (u_H, P_H) on S^3_R satisfies*

$$\nabla_{u_H} u_H = -\operatorname{grad} P_H, \qquad \partial_t u_H + \nu \Delta_H u_H = f_H, \tag{30}$$

for some solenoidal f_H. Put $c = 4\nu/R^2$ and $s(t) = e^{ct}$. Then

$$\boxed{u_{\mathrm{EM}} = s\, u_H, \quad P_{\mathrm{EM}} = s^2 P_H, \quad f_{\mathrm{EM}} = s\, f_H} \tag{31}$$

solves the corresponding incompressible Navier–Stokes equation with A_{EM} in place of Δ_H. The result applies to each of Secs. 2 and 3 with the pressure balanced as above; it does not apply in general to solutions whose quadratic convection has a non-gradient part.

Proof. Since s depends only on time, $\nabla_{s u_H}(s u_H) = s^2 \nabla_{u_H} u_H = -\operatorname{grad}(s^2 P_H)$. Further,

$$\partial_t(s u_H) + \nu A_{\mathrm{EM}}(s u_H) = s\big(\partial_t u_H + \nu \Delta_H u_H\big) + (s' - cs) u_H = s f_H.$$

The solenoidal constraint is unchanged. This proves the claim. □

Corollary 4.2 (Spectral decay, pressure beats and persistent Killing field)**.** *In the unforced toroidal family of Sec. 2, replace every rate γ_n with*

$$\boxed{\gamma_n^{\mathrm{EM}} = \frac{4\nu n(n+2)}{R^2} = \gamma_n^H - \frac{4\nu}{R^2}.} \tag{32}$$

The pressure remains Eq. (11) with these new rates; its nonlinear beat exponents become $\gamma_n^{\mathrm{EM}} + \gamma_k^{\mathrm{EM}}$. The $n = 0$ components are preserved, and, for a finite modal sum,

$$u_{\mathrm{EM}}(t) \longrightarrow u_\infty = a_0\partial_\phi + b_0\partial_\psi, \quad \|u_{\mathrm{EM}}(t) - u_\infty\|_2 \le e^{-12\nu t/R^2}\|u(0) - u_\infty\|_2. \tag{33}$$

Thus the Hodge and Ebin–Marsden models have the same exact toroidal mode shapes but different asymptotic dynamics.

Proof. The squared curl eigenvalues are $\kappa_n^2 = 4(n+1)^2/R^2$. Subtracting $4/R^2$ gives Eq. (32). All Jacobi modes are L^2-orthogonal in their respective angular components. For $n \ge 1$, $n(n+2) \ge 3$, giving the stated bound. □

Proposition 4.3 (Six exact Killing-momentum balance laws)**.** *For any smooth divergence-free solution, not only those in the special sectors above, and each Killing field K on the round S^3_R, define $Q_K = \int g_R(u, K)\,\mathrm{d}V$. Then*

$$\boxed{\begin{aligned} \frac{\mathrm{d}Q_K^H}{\mathrm{d}t} &= -\frac{4\nu}{R^2}Q_K^H + \int_{S^3_R} g_R(f_H, K)\,\mathrm{d}V, \\ \frac{\mathrm{d}Q_K^{\mathrm{EM}}}{\mathrm{d}t} &= \int_{S^3_R} g_R(f_{\mathrm{EM}}, K)\,\mathrm{d}V. \end{aligned}} \tag{34}$$

The six-dimensional real Killing space has two $\mathfrak{su}(2)$ factors. Thus without forcing all six charges decay for Hodge viscosity, but are conserved for Ebin–Marsden viscosity. This is a concrete instance of the known Killing-field conservation mechanism for deformation-tensor viscosity; no claim of historical priority is made.

Proof. Using compactness and $\operatorname{div} u = 0$, integration by parts gives

$$\int g_R(\nabla_u u, K)\,\mathrm{d}V = -\int u^i u^j \nabla_i K_j\,\mathrm{d}V = 0$$

by the Killing equation. The pressure term also pairs to zero. The positive Hodge Laplacian is self-adjoint and $\Delta_H K = 4K/R^2$ on the round sphere, whereas $A_{\mathrm{EM}}K = 0$. Pairing the two equations with K proves the result. □

5 Duration-optimized projective forcing

Both viscosity conventions admit exactly the fixed-degree Hopf/Beltrami control construction of Sec. 3, with decay parameter

$$\gamma_m^H = \frac{\nu(m+2)^2}{R^2}, \qquad \gamma_m^{\mathrm{EM}} = \frac{\nu m(m+4)}{R^2}. \tag{35}$$

This follows immediately from $\Delta_H U = \lambda_m^2 U$ and Eq. (29). All Berry and pressure-projector identities remain identical except for the scalar radial damping coefficient.

Corollary 5.1 (Optimal cycle time and minimal integrated forcing)**.** *For a specified projective loop of constant spatial polynomial degree $m \ge 1$, constant coefficient radius $r > 0$, Fubini–Study length L_{FS}, and Berry holonomy β, put*

$$D = \sqrt{L_{\mathrm{FS}}^2 + \operatorname{dist}(\beta, 2\pi\mathbb{Z})^2}.$$

If the loop is nontrivial ($D > 0$) and one can choose its time parameterization and physical lift, then among all cycle durations $T > 0$ for which the full velocity closes,

$$\boxed{T_* = \frac{D}{\gamma}, \qquad \min_{T>0}\int_0^T \|f(t)\|_{L^2}^2\,\mathrm{d}t = 2C_m r^2 \gamma D,} \tag{36}$$

where γ is either positive rate in Eq. (35). No assertion is made about changing the projective curve itself, or unrestricted actuator geometry.

Proof. Reparameterize the fixed projective curve at constant Fubini–Study speed and choose the physical phase with constant minimum-magnitude vertical connection velocity. Then Theorem 3.2 is saturated:

$$\mathcal{A}(T) = C_m r^2(\gamma^2 T + D^2/T).$$

By the arithmetic–geometric mean inequality (or differentiation), $\mathcal{A}(T) \geq 2C_m r^2 \gamma D$, with equality at $T = D/\gamma$. □

Corollary 5.2 (Exact viscosity contrast for the quadratic braid)**.** *For the degree-two braid in Eq. (27), $r = 1$ and $D = \pi\sqrt{2}$. Consequently,*

$$\begin{aligned} T_*^H &= \frac{\pi\sqrt{2}R^2}{16\nu}, \qquad T_*^{\mathrm{EM}} = \frac{\pi\sqrt{2}R^2}{12\nu}, \\ \frac{\mathcal{A}_{\min}^H}{\mathcal{A}_{\min}^{\mathrm{EM}}} &= \frac{16}{12} = \frac{4}{3}, \qquad \mathcal{A}_{\min} = 2\pi\sqrt{2}C_2\gamma. \end{aligned} \tag{37}$$

The ratio distinguishes two specified curved-space viscous models; it is not a universal physical constant or a measured property of a fluid.

Limitations. The curvature-shift conjugacy does not identify the Hopf projective sector with the Jacobi toroidal sector and does not control non-invariant perturbations. Its Noether component is known in the geometric fluid literature. The time-optimal forcing calculation is a constrained variational statement, not a solution of a quantum isoholonomic speed-limit problem and not a cosmological prediction.

6 An exact hybrid Hopf–Jacobi interaction identity

The preceding two invariant families are not generally closed under addition. Nevertheless, they admit an exact *controlled* superposition theorem. We use the L^2 Leray projector $\mathbb{P}$ onto divergence-free vector fields on S_R^3. By Φ_q we mean the unique zero-mean solution of $\Delta_{\mathrm{geom}}\Phi_q = \operatorname{div} q$, with $q = \mathbb{P}q + \nabla\Phi_q$; here $\Delta_{\mathrm{geom}} = \operatorname{div}\operatorname{grad}$ is the nonpositive scalar Laplacian. No boundary conditions are needed on the closed sphere.

Theorem 6.1 (Exact hybrid forcing and nonlinear leakage)**.** *Let $v(t)$ be an arbitrary smooth doubly rotational field of Eq. (5), and let $w(t) = \mathscr{U}[p(t)]$ belong to a fixed positive-curl Hopf eigenspace, $\operatorname{curl} w = \lambda_m w$, $\lambda_m = (m+2)/R$. Choose either $\mathcal{A} = \Delta_H$ (Hodge) or $\mathcal{A} = \Delta_H - 4/R^2$ (Ebin–Marsden). Write*

$$F_v = \partial_t v + \nu\mathcal{A}v, \qquad F_w = \partial_t w + \nu\mathcal{A}w. \tag{38}$$

Then $U = v + w$ is an exact *solution of incompressible Navier–Stokes with solenoidal force*

$$\boxed{F = F_v + F_w + \mathbb{P}\big[w \times (\lambda_m v - \operatorname{curl} v)\big]} \tag{39}$$

and pressure

$$P = P_v + P_w - v \cdot w - \Phi_q, \quad q = w \times (\lambda_m v - \operatorname{curl} v), \tag{40}$$

where $\nabla_v v = -\operatorname{grad} P_v$ and $P_w = -|w|^2/2$. The identity holds for any differentiable coefficient trajectories and either viscosity convention. It does not *assert that the uncontrolled sum solves the equations.*

Proof. On every oriented Riemannian three-manifold, the cross-product identity gives

$$\nabla_v w + \nabla_w v = \operatorname{grad}(v \cdot w) - v \times \operatorname{curl} w - w \times \operatorname{curl} v = \operatorname{grad}(v \cdot w) + q.$$

The two self-advection terms are gradients by the toroidal and Beltrami hypotheses. The time derivative and the selected viscous operator are linear. Splitting $q = \mathbb{P}q + \operatorname{grad} \Phi_q$ and collecting gradient terms proves Eqs. (39)–(40). Since the cross product, Leray projection, and coefficient paths are smooth, the result has the stated regularity. □

Proposition 6.2 (Noether selection for hybrid leakage)**.** *For every one of the six Killing fields K of the round S^3_R,*

$$\boxed{\int_{S^3_R} \langle \mathbb{P}q, K \rangle \, dV = 0.} \tag{41}$$

Thus the additional force in Eq. (39) cannot directly inject a Killing momentum. It may, however, excite other solenoidal modes.

Proof. The Killing equation makes ∇K antisymmetric. Integration by parts, using $\div v = \div w = 0$, yields

$$\int \langle \nabla_v w + \nabla_w v, K \rangle \, dV = -\int (v^i w^j + w^i v^j) \nabla_i K_j \, dV = 0.$$

The integral of K against any scalar gradient vanishes. Using the cross-term identity and orthogonality of the Leray decomposition gives the result. □

7 Spectral detuning, an arithmetic resonance, and a forced-action bound

Define $V_{n,+} = (U^A_n + U^B_n)/\sqrt{2}$, $V_{n,-} = (U^A_n - U^B_n)/\sqrt{2}$. Then $\operatorname{curl} V_{n,\sigma} = \sigma \kappa_n V_{n,\sigma}$ and $\|V_{n,\sigma}\|_2^2 = d_n$, where $d_n = \pi^2 R^5/(n+1)$. Write $v = \sum_{n,\sigma} c_{n,\sigma} V_{n,\sigma}$ with finitely many real coefficients. The spectral detuning functional at λ is

$$\mathfrak{D}_\lambda(v)^2 = \|(\operatorname{curl} - \lambda) v\|_2^2 = \sum_{n,\sigma} d_n |c_{n,\sigma}|^2 (\sigma \kappa_n - \lambda)^2. \tag{42}$$

It is a nonnegative spectral diagnostic, *not* a new topological invariant or a conserved charge in viscous dynamics.

Theorem 7.1 (Sharp pointwise polynomial envelope and detuning estimate)**.** *For the normalized coefficients $b_k = c_k/\sqrt{\binom{m}{k}}$ of the fixed-degree polynomial p, one has $\|w\|_{L^\infty} \le \|b\|_{\mathbb{C}^{m+1}}$. Consequently the* additional *projected nonlinear force satisfies*

$$\boxed{\|\mathbb{P}q\|_2 \le \|b\| \, \mathfrak{D}_{\lambda_m}(v).} \tag{43}$$

For unforced radial Hodge (respectively Ebin–Marsden) coefficient evolution $b(t) = e^{-\Gamma_m t} b(0)$ and $c_{n,\sigma}(t) = e^{-\gamma_n t} c_{n,\sigma}(0)$, this entails the finite total interaction-cost estimate

$$\int_0^\infty \|\mathbb{P}q(t)\|_2^2 \, dt \le \|b(0)\|^2 \sum_{n,\sigma} \frac{d_n |c_{n,\sigma}(0)|^2 (\sigma \kappa_n - \lambda_m)^2}{2(\Gamma_m + \gamma_n)}. \tag{44}$$

All terms with vanishing numerator are interpreted as zero; in the present $m \ge 1$ case $\Gamma_m > 0$ under either convention.

Proof. The globally orthonormal frame gives $|w| = |p|$. The finite-dimensional Cauchy–Schwarz inequality yields

$$|p(z)|^2 \le \|b\|^2 \sum_{k=0}^{m} \binom{m}{k} |z_1|^{2(m-k)} |z_2|^{2k} = \|b\|^2.$$

As the Leray projection is an orthogonal contraction on L^2, $\|\mathbb{P}q\|_2 \le \|w\|_\infty \|(\lambda_m - \operatorname{curl})v\|_2$. The signed Jacobi modes are mutually orthogonal and have the displayed norms. Substitution of the exponential solutions, squaring and integrating term by term gives Eq. (44). □

Corollary 7.2 (Exact even-degree Hopf–Jacobi resonance)**.** *A positive-curl Hopf polynomial field of degree $m \ge 1$ shares a signed toroidal Jacobi curl eigenvalue if and only if m is even. In that case $n = m/2$ and $\sigma = +$, since*

$$\lambda_m = \frac{m+2}{R} = \frac{2(n+1)}{R} = \kappa_n. \tag{45}$$

For any superposition of positive-curl fields in that common eigenspace, the additional hybrid forcing $\mathbb{P}q$ vanishes, and the unforced radial decay of the sum is an exact full nonlinear solution. For odd m, the spectral detuning *satisfies*

$$\mathfrak{D}_{\lambda_m}(v) \ge R^{-1} \|v\|_2 \tag{46}$$

for every toroidal finite expansion. This inequality does not *imply a nonzero lower bound on the projected force in Eq. (43): Leray projection can annihilate special cross terms.*

Proof. The positive toroidal curl values are even positive multiples of R^{-1}, while $\lambda_m R = m + 2$ is an integer. Equality is equivalent to $m = 2n$. At equality $\operatorname{curl}(v + w) = \lambda_m(v + w)$, so its full nonlinear acceleration is the gradient $\operatorname{grad}(|v + w|^2/2)$. When m is odd, every difference $|\sigma\kappa_n - \lambda_m| \ge R^{-1}$, and Eq. (42) gives the claimed estimate. □

8 Exact spectral-variance diagnostic for the hybrid family

The interaction term in Theorem 6.1 is controlled by spectral detuning, but the latter does not measure the true Leray-projected forcing from below. A different, fully observable quantity characterizes the *Beltrami compatibility locus* without making such an incorrect inference. For any nonzero smooth solenoidal U write

$$E(U) = \tfrac{1}{2}\|U\|_2^2, \quad H(U) = \langle U, \operatorname{curl} U\rangle_2, \quad Z(U) = \|\operatorname{curl} U\|_2^2, \quad \mathscr{V}(U) = Z(U) - \frac{H(U)^2}{2E(U)}. \tag{47}$$

Theorem 8.1 (Helicity-defect identity and exact two-frequency separation)**.** *On a closed oriented Riemannian three-manifold, every nonzero smooth divergence-free U satisfies*

$$\boxed{\mathscr{V}(U) = \left\| \operatorname{curl} U - \frac{H(U)}{2E(U)} U \right\|_2^2 \ge 0.} \tag{48}$$

Equality holds if and only if U is a curl eigenfield. On the round S^3_R, take $U = v + w$, where v is a single signed Jacobi eigenfield with $\operatorname{curl} v = \kappa v$, w a fixed-degree Hopf polynomial eigenfield with $\operatorname{curl} w = \lambda w$, and $\kappa \ne \lambda$. Then the two fields are L^2-orthogonal and, setting $A = \|v\|_2^2 > 0$, $B = \|w\|_2^2 > 0$,

$$\boxed{\mathscr{V}(U) = \frac{AB}{A+B}(\kappa - \lambda)^2 > 0.} \tag{49}$$

In contrast, if $\kappa = \lambda$ then U is itself a curl eigenfield and $\mathscr{V}(U) = 0$, even though $\langle v, w\rangle$ need not vanish. For separate radial viscous attenuation, $A(t) = A_0 e^{-2\gamma_v t}$ and $B(t) = B_0 e^{-2\gamma_w t}$ in the nonresonant case, so Eq. (49) gives the exact time-resolved measurable defect.

Proof. Complete the square in the L^2 Hilbert space: $\|\operatorname{curl} U - \mu U\|_2^2 = Z - 2\mu H + 2\mu^2 E$; the minimizing real scalar is $\mu = H/(2E)$. Equality means $\operatorname{curl} U = \mu U$ pointwise for smooth fields. On the closed three-sphere curl is self-adjoint on the divergence-free domain, so eigenfields of distinct eigenvalues are orthogonal. Thus $2E = A + B$, $H = \kappa A + \lambda B$, $Z = \kappa^2 A + \lambda^2 B$, and substitution gives Eq. (49). When the eigenvalues coincide, linearity of curl proves the last equality without requiring orthogonality. Exponential attenuation follows from the Hodge or Ebin–Marsden spectral rates already established. □

Corollary 8.2 (A dimensionless spectral coherence observable)**.** *For $Z(U) > 0$, the ratio*

$$\boxed{\chi(U) = \frac{|H(U)|}{\sqrt{2E(U)Z(U)}} \in [0,1]} \tag{50}$$

measures the degree of spectral alignment, with $\chi = 1$ exactly for nonzero curl eigenfields. For two distinct nonzero modes the spectral defect $\mathscr{V}$ is positive, hence $\chi < 1$ even if the signed-curl eigenvalues are close. The quantity $\mathscr{V}$ is not *asserted to be conserved: only its exact zero locus is preserved by common-eigenvalue radial evolution. This observable is a direct consequence of the classical Cauchy–Schwarz inequality; any priority claim concerns its integration into the specified Hopf–Jacobi construction, not the underlying inequality.*

Interpretation and limits. The spatial zeros of p_m visualize the Hopf component w; when v is added they need not be zeros of $U = v + w$. Streamlines can be obtained by integrating the explicit four-dimensional tangent vector field and then applying stereographic projection. That geometric computation is separate from the Leray projection appearing in the force. A vanishing variance implies a single Beltrami eigenvalue but does not, in arbitrary fields, establish novel algebraic topology, a physical knot experiment, or global Navier–Stokes regularity.

9 Commutation-protected nonresonant closure and a diagnostic counterexample

The previous even-degree Hopf–Jacobi matching condition is sufficient for exact unforced nonlinear closure, but it is *not necessary*. An exact second mechanism is supplied by commuting solenoidal curl eigenfields. In particular, spectral variance measures departure from a single curl eigenspace; it need not measure the nonlinear force. This distinction changes which conclusions the HCC phase-space diagnostic may legitimately display.

Theorem 9.1 (Leray–Lie bracket criterion on a homology three-sphere)**.** *Let (M^3, g) be a closed oriented Riemannian manifold with $H^1_{\mathrm{dR}}(M) = 0$. Let u, v be smooth real divergence-free fields satisfying*

$$\operatorname{curl} u = \lambda u, \qquad \operatorname{curl} v = \kappa v, \tag{51}$$

with real constant signed eigenvalues. If $\mathbb{P}$ is the L^2 Leray projector, then

$$\mathbb{P}(\nabla_u v + \nabla_v u) = (\lambda - \kappa)\mathbb{P}(u \times v), \tag{52}$$

$$\operatorname{curl}(u \times v) = -[u, v]. \tag{53}$$

Here $[u, v] = \nabla_u v - \nabla_v u$ is the Lie bracket. Consequently, if $\lambda \neq \kappa$,

$$\boxed{\mathbb{P}(\nabla_u v + \nabla_v u) = 0 \iff [u, v] = 0.} \tag{54}$$

When $\lambda = \kappa$, the left-hand side vanishes regardless of the bracket.

Proof. The metric cross-product identity gives

$$\nabla_u v + \nabla_v u = \operatorname{grad}\langle u, v\rangle - u \times \operatorname{curl} v - v \times \operatorname{curl} u = \operatorname{grad}\langle u, v\rangle + (\lambda - \kappa) u \times v. \tag{55}$$

The first identity follows by applying $\mathbb{P}$. On a three-manifold, $\operatorname{curl}(u\times v) = \nabla_v u - \nabla_u v + u \operatorname{div} v - v \operatorname{div} u = -[u, v]$, proving the second. For $\lambda \neq \kappa$, the projected mixed term vanishes if and only if $u \times v$ lies in the gradient subspace. If its curl vanishes, the metric-dual one-form is closed; the hypothesis $H^1_{\mathrm{dR}}(M) = 0$ makes it exact. Conversely every gradient has vanishing curl. The equal-eigenvalue case is immediate from the first identity. □

Corollary 9.2 (An exact unforced two-rate Navier–Stokes family)**.** *Under the hypotheses of Theorem 9.1, assume additionally* $[u, v] = 0$ *and that* $\mathcal{A}u = \rho_\lambda u$, $\mathcal{A}v = \rho_\kappa v$, *with* $\mathcal{A} = \Delta_H$ *or (on round* S^3_R*)* $\mathcal{A} = \Delta_H - 4R^{-2} I$. *Then*

$$U(t) = a\, e^{-\nu\rho_\lambda t} u + b\, e^{-\nu\rho_\kappa t} v \tag{56}$$

is an exact smooth unforced incompressible Navier–Stokes solution for $t \geq 0$. *For the Hodge operator* $\rho_\lambda = \lambda^2$ *and* $\rho_\kappa = \kappa^2$. *For any time* t, *let* Φ *be the zero-mean solution* $\operatorname{grad}\Phi = u \times v$, *which exists on* M *because the fields commute. A compatible pressure is*

$$P(t) = -\frac{1}{2}|U(t)|^2 - (\lambda - \kappa) abe^{-\nu(\rho_\lambda + \rho_\kappa)t}\Phi, \tag{57}$$

up to an arbitrary function of time.

Proof. Theorem 9.1 shows that the cross acceleration is a gradient, while each diagonal Beltrami self-acceleration is a gradient. The exact Φ exists by Eq. (53) and vanishing H^1. The time derivative in Eq. (56) cancels the spectral viscosity term and Eq. (57) cancels the full remaining gradient. Divergence vanishes by linearity. □

9.1 A fixed-Hopf family with a commuting opposite-chirality Killing field

Set $z_1 = a + ib$, $z_2 = c + id$ and use the quaternion frame of Eq. (20). Define the globally smooth Killing fields

$$K_+ = \partial_\phi + \partial_\psi, \qquad K_- = \partial_\phi - \partial_\psi, \tag{58}$$

with the orientation of Lemma 2.1, so that $\operatorname{curl} K_+ = 2K_+/R$ and $\operatorname{curl} K_- = -2K_-/R$. Write $Z = X_2 + iX_3$ for the fixed-Hopf complex null frame. Direct differentiation of the displayed real four-vector fields yields

$$[K_+, Z] = 2iZ, \qquad [K_-, Z] = 0, \tag{59}$$

where $[\cdot,\cdot]$ acts complex-linearly on Z. Thus for homogeneous $p(z_1, z_2) = \sum_{j=0}^m c_j z_1^{m-j} z_2^j$,

$$[K_+, \mathscr{U}[p]] = \operatorname{Re}(i(m+2)pZ), \tag{60}$$

$$[K_-, \mathscr{U}[p]] = \operatorname{Re}\left(i \sum_{j=0}^m (m-2j) c_j z_1^{m-j} z_2^j Z \right). \tag{61}$$

All commutators use real vector fields; the real-part notation applies only to their coefficient representation.

Theorem 9.3 (Complete fixed-degree K_- centralizer in the fixed-Hopf polynomial sector)**.** *For a nonzero homogeneous holomorphic polynomial* p *of degree* $m \geq 0$, *let* $w_p = \mathscr{U}[p] = \operatorname{Re}(pZ)$. *Then*

$$\boxed{[K_-, w_p] = 0 \iff m = 2k \text{ and } p(z_1, z_2) = c(z_1 z_2)^k,\ c \in \mathbb{C} \setminus \{0\}.} \tag{62}$$

Moreover $[K_+, w_p] \neq 0$ *for every nonzero* p *of degree* $m \geq 0$.

Proof. The Hopf frame X_2, X_3 is real-linearly independent at each point, so $\mathrm{Re}(qZ) = 0$ as a real field if and only if the complex coefficient q vanishes. In Eq. (61), the monomials are linearly independent on S^3 (homogeneity extends their identity to $\mathbb{C}^2$), hence $(m-2j)c_j = 0$ for every j. A nonzero coefficient is possible only when $m = 2j$, and then $p = c(z_1 z_2)^j$. Conversely such a monomial has zero K_- weight. Eq. (60) has strictly positive weight $m+2$, so the K_+ commutator cannot vanish identically. □

Corollary 9.4 (Explicit nonresonant, multi-decay exact solutions outside the toroidal subspace)**.** *For any integer $k \geq 1$, take $p_k = c(z_1 z_2)^k$, $c \neq 0$, and $w_k = \mathrm{Re}(p_k Z)$; then*

$$\operatorname{curl} w_k = \frac{2k+2}{R} w_k, \qquad \operatorname{curl} K_- = -\frac{2}{R} K_-, \qquad [K_-, w_k] = 0. \tag{63}$$

Consequently, for arbitrary real a, b, the following is an exact unforced *Hodge-viscous solution:*

$$\boxed{U_k(t) = a e^{-4\nu t/R^2} K_- + b e^{-(2k+2)^2 \nu t/R^2} w_k.} \tag{64}$$

The Ebin–Marsden convention instead fixes the Killing component and damps the polynomial part with rate $\nu[(2k+2)^2 - 4]/R^2$. The two components have different signed curl eigenvalues and different Hodge damping rates*; their nonlinear Leray forcing vanishes identically. The field w_1 has a nonzero x-component on an open set, so this family is not confined to the previously analyzed doubly rotational toroidal sector.*

Proof. The polynomial curl eigenidentity is Eq. (22) and the curl construction preceding it; the commutation follows from Theorem 9.3. Apply Corollary 9.2. For w_1, in ambient real coordinates the derivative of $x = c^2 + d^2$ is nonzero as a polynomial on S^3, for example at $q = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ it equals $-\frac{1}{2}$ with the specified frame and $p = z_1 z_2$, so the field is not toroidal. □

Proposition 9.5 (Closed polynomial pressure without Poisson inversion)**.** *With the notation of Corollary 9.4, let $a(t) = a e^{-4\nu t/R^2}$ and $b(t) = b e^{-(2k+2)^2 \nu t/R^2}$ in the Hodge convention, and set $s = z_1 z_2$. The* exact *cross-product primitive is*

$$\boxed{K_- \times_{g_R} w_k = \frac{R^2}{k+1} \operatorname{grad}_{g_R} \mathrm{Im}(c s^{k+1}).} \tag{65}$$

Consequently a globally regular pressure, requiring no inversion of the scalar Laplacian, is

$$\boxed{P_k(t,q) = -\frac{1}{2}|U_k(t,q)|^2_{g_R} + \frac{(2k+4)R}{k+1} a(t) b(t) \,\mathrm{Im}[c(z_1 z_2)^{k+1}].} \tag{66}$$

Both identities extend smoothly through the collapsed Clifford circles.

Proof. At unit radius the quaternion frame identities yield $K_- \times X_2 = \operatorname{grad} \mathrm{Im}(z_1 z_2)$ and $K_- \times X_3 = -\operatorname{grad} \mathrm{Re}(z_1 z_2)$. Writing $w_k = \mathrm{Re}(c s^k) X_2 - \mathrm{Im}(c s^k) X_3$, we obtain

$$K_- \times w_k = \mathrm{Re}(c s^k) \operatorname{grad} \mathrm{Im}\, s + \mathrm{Im}(c s^k) \operatorname{grad} \mathrm{Re}\, s = \frac{1}{k+1} \operatorname{grad} \mathrm{Im}(c s^{k+1}).$$

Rescaling $g_R = R^2 g_1$ and the orthonormal Hopf vectors $X_i^{(R)} = R^{-1} X_i^{(1)}$ gives the factor R^2 in Eq. (65). Finally, $\nabla_{U_k} U_k = \operatorname{grad}(|U_k|^2/2) + a(t)b(t)(\lambda_{K_-} - \lambda_{w_k}) K_- \times w_k$ with $\lambda_{K_-} - \lambda_{w_k} = -(2k+4)/R$. Substituting Eq. (65) yields $\nabla_{U_k} U_k = -\operatorname{grad} P_k$. Equation (66) follows. □

9.2 Strictly positive helicity variance with *zero* projected nonlinear force

The two signed curl eigenvalues in Eq. (64) are distinct, so K_- and w_k are L^2 orthogonal. Direct integration gives

$$N_K = \|K_-\|_2^2 = 2\pi^2 R^5, \qquad N_w = \|w_k\|_2^2 = 2\pi^2 R^3 |c|^2 \frac{(k!)^2}{(2k+1)!}. \tag{67}$$

For nonzero a, b, let $A(t) = a^2 N_K e^{-8\nu t/R^2}$ and $B(t) = b^2 N_w e^{-2(2k+2)^2 \nu t/R^2}$. The exact energy, helicity and enstrophy are

$$\mathcal{E}(t) = \tfrac{1}{2}(A + B), \tag{68}$$
$$\mathcal{H}(t) = \tfrac{1}{R}[-2A + (2k+2)B], \tag{69}$$
$$\mathcal{Z}(t) = \tfrac{1}{R^2}[4A + (2k+2)^2 B]. \tag{70}$$

In contrast with the vanishing nonlinear projected force, the spectral helicity deficit is

$$\boxed{\mathscr{V}(U_k(t)) = \frac{A(t)B(t)}{A(t)+B(t)} \frac{(2k+4)^2}{R^2} > 0, \qquad \mathbb{P}(\nabla_{U_k} U_k) = 0.} \tag{71}$$

The equality and inequality hold for each finite t with nonzero initial amplitudes. This gives a concrete counterexample to an invalid inference *positive spectral variance implies nonzero nonlinear leakage*. The previously established resonance criterion remains sufficient for compatibility but is not necessary.

Scientific provenance and relation to the external catalogue. The cross-product identity, Poincaré lemma on S^3, and spectral variance are classical. Exact viscous Beltrami flows have a substantial earlier literature [15, 16]. This section identifies a reproducible *joint specialization* within the cited fixed-Hopf polynomial family. The historical novelty of this specialization has not been established by a global prior-art audit. OpenAI Math result family 376 treats *forced flat* fluid computations and supplies no theorem that would imply general unforced regularity on a curved sphere. OpenAI Math result family 350 concerns nodal sets for other metric classes; it does not prove stability of the fixed-round Hopf centralizer under metric perturbation. None of these statements establishes a physical cosmic S^3 topology.

10 Exact finite-time minimum-energy control within each reducing family

The possibility of prescribing arbitrary modal trajectories makes exact optimal-control statements possible *within each family separately*. They should not be silently added across the hybrid family, which has the extra cross forcing of Theorem 6.1.

Theorem 10.1 (Endpoint control Gramian)**.** *Let $c : [0,T] \to \mathbb{R}^d$ (or $\mathbb{C}^d$) solve $\dot{c} + \gamma c = h(t)$ with fixed $c(0) = c_0$, $c(T) = c_T$ and $\gamma \geq 0$. For any Hilbert-space weight $D > 0$,*

$$\boxed{\min_h \int_0^T D\|h(t)\|^2 dt = \frac{D\|c_T - e^{-\gamma T} c_0\|^2}{W_\gamma(T)}, \quad W_\gamma(T) = \begin{cases} \frac{1-e^{-2\gamma T}}{2\gamma}, & \gamma > 0, \\ T, & \gamma = 0. \end{cases}} \tag{72}$$

Equality is achieved by $h_*(t) = e^{-\gamma(T-t)}(c_T - e^{-\gamma T} c_0)/W_\gamma(T)$. In the toroidal sector the full forcing cost is the sum over independent signed modes with $D = d_n$ and $\gamma = \gamma_n$; in the polynomial Hopf sector $D = C_m$, $d = m+1$ complex coefficients and $\gamma = \Gamma_m$. The cost is for all freely accessible solenoidal coefficients, not a restricted physical actuator geometry.

Proof. Duhamel's formula reads $c_T - e^{-\gamma T}c_0 = \int_0^T e^{-\gamma(T-t)}h(t)dt$. Apply Cauchy–Schwarz and compute the squared kernel norm $W_\gamma(T)$. Equality is attained precisely for h proportional to the kernel. Modal orthogonality and the isometric polynomial coefficient map supply the weights. □

11 Finite spectral tomography and its hard limits

For an unforced Jacobi solution, the energy and helicity contain only distinct decay nodes $q_n = e^{-2\gamma_n \Delta}$ when $\nu, R, \Delta > 0$ are known. The two scalar observations are complementary to the *pressure* measurements, whose mixed sums need not be unique.

Theorem 11.1 (Modewise energy–helicity reconstruction)**.** *Fix N, $\Delta > 0$, $\nu > 0$, $R > 0$ and either viscosity operator. Given* exact noiseless *values $(\mathcal{E}(j\Delta), \mathcal{H}(j\Delta))$ for $j = 0, \dots, N$, one uniquely reconstructs, for every $n = 0, \dots, N$,*

$$S_n = a_n^2 + b_n^2, \qquad P_n = a_n b_n.$$

More precisely, invert the known Vandermonde matrix $V_{jn} = q_n^j$ to obtain the mode amplitudes e_n, h_n of energy and helicity; then

$$S_n = \frac{2(n+1)}{\pi^2 R^5} e_n, \qquad P_n = +\frac{h_n}{4\pi^2 R^4}, \qquad \{a_n^2, b_n^2\} = \left\{ \frac{S_n \pm \sqrt{S_n^2 - 4P_n^2}}{2} \right\}. \tag{73}$$

The sign of $a_n b_n$ is recovered, but the global sign $(a_n, b_n) \mapsto (-a_n, -b_n)$ and their exchange are unresolved in general. Modewise consistency requires $S_n^2 \geq 4P_n^2$, equivalently $|h_n| \leq 2\kappa_n e_n$. This is a finite-dimensional uniqueness result, not *a stability bound for noisy measurements.*

Proof. The decay nodes are strictly distinct because γ_n increases with n for both viscosity conventions. The determinant of the square Vandermonde matrix is $\prod_{j<k}(q_k - q_j) \neq 0$. The explicit energy and helicity identities of Corollary 2.3 apply at every time with the chosen rates; inversion yields e_n, h_n. Solving the quadratic equation with roots a_n^2, b_n^2 establishes Eq. (73). The final inequality is $2|a_n b_n| \leq a_n^2 + b_n^2$. □

Remark 11.2 (Distinguishing phase, topology, and observability)**.** *The scalar values $\mathcal{E}, \mathcal{H}$ do not determine general velocity fields; their sufficiency above depends on the complete finite Jacobi ansatz and known modal nodes. Nor is the coefficient-space Berry phase a newly discovered quantum-gravitational holonomy. The braid exchanges zeros of a polynomial: these need not be material fluid-particle trajectories. No observation here establishes global S^3 cosmic topology.*

12 A dynamical spectral–statistical–projective identity

The preceding variance diagnostic is formed from the *signed curl* generator. A different, physically time-resolved projective observable comes from the squared-curl viscosity generator. This distinction is essential: equal Hodge frequencies need not imply equal helicity or equal signed curl eigenvalues.

Let $\{e_j\}_{j=1}^N$ be a fixed orthonormal real basis of smooth divergence-free eigenfields of curl on the round S_R^3, $\operatorname{curl} e_j = \lambda_j e_j$. Define a finite spectral history

$$U(t) = \sum_{j=1}^{N} c_j e^{-\nu(\lambda_j^2 - c_R)t} e_j, \quad c_R = 0 \text{ (Hodge)}, \quad c_R = 4/R^2 \text{ (Ebin–Marsden)}. \tag{74}$$

We consider this history as a *Navier–Stokes solution only if* its nonlinear Leray projection vanishes at every time. This condition holds for the commuting family of Corollary 9.4 and for the pressure-balanced

toroidal sector, but not for arbitrary choices of the e_j. Regardless of PDE closure, Eq. (74) is a valid spectral heat orbit.

Set $W(t) = \|U(t)\|_2^2 > 0$, $M_{2r}(t) = \sum_j \lambda_j^{2r} c_j^2 e^{-2\nu(\lambda_j^2 - c_R)t}$ for $r = 1, 2$; in particular $M_2 = \|\operatorname{curl} U\|_2^2$ and $M_4 = \|\Delta_H U\|_2^2$. Let $[U(t)]$ denote its ray in the real projective Hilbert space, embedded in the complex Fubini–Study space with metric convention $d_{\mathrm{FS}}([x],[y]) = \arccos|\langle \widehat{x}, \widehat{y}\rangle|$.

Theorem 12.1 (Viscous projective-speed / logarithmic-energy curvature identity)**.** *For every nonzero history* (74) *with* $\nu > 0$*, the physical-time projective speed is exactly*

$$\boxed{v_{\mathrm{FS}}(t)^2 = \nu^2 \left(\frac{M_4}{W} - \frac{M_2^2}{W^2} \right) = \frac{1}{4} \frac{\mathrm{d}^2}{\mathrm{d}t^2} \log W(t) = \frac{\nu^2}{W^2} \sum_{i<j} A_i(t) A_j(t) (\lambda_i^2 - \lambda_j^2)^2,} \tag{75}$$

where $A_j(t) = c_j^2 e^{-2\nu(\lambda_j^2 - c_R)t}$*. The mean Hodge frequency* $\overline{\lambda^2} = M_2/W$ *satisfies*

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t} \frac{M_2}{W} = -2\nu \left(\frac{M_4}{W} - \frac{M_2^2}{W^2} \right) = -\frac{2}{\nu} v_{\mathrm{FS}}^2 \le 0.} \tag{76}$$

The speed vanishes exactly when all nonzero modes have the same absolute *curl eigenvalue* $|\lambda_j|$*; their signed eigenvalues may differ. This is a statement about the stated linear reducing family, not the unrestricted Navier–Stokes phase flow.*

Proof. Write $\psi(t) = U(t)/\sqrt{W(t)}$ and $\mathcal{A} = \Delta_H - c_R I$. The linear equation $\dot{U} = -\nu \mathcal{A} U$ implies $\dot{\psi} = -\nu(\mathcal{A} - \langle \mathcal{A} \rangle_\psi)\psi$, which is orthogonal to ψ. Hence the Fubini–Study speed is $\|\dot{\psi}\|^2 = \nu^2 \mathrm{Var}_\psi(\mathcal{A})$. Shifting an operator by a scalar leaves its variance unchanged, giving the first expression in (75); expanding the weighted variance gives the pairwise expression. Directly, $\dot{W} = -2\nu(M_2 - c_R W)$ and $\ddot{W} = 4\nu^2(M_4 - 2c_R M_2 + c_R^2 W)$, so $(\log W)'' = 4\nu^2(M_4/W - M_2^2/W^2)$. Quotient differentiation gives (76). All weights are nonnegative, so equality holds precisely on a single λ^2 eigenshell. □

Corollary 12.2 (Classical Fisher information and viscosity-independent projective shape)**.** *The mode-energy proportions* $q_j(t) = A_j(t)/W(t)$ *constitute a probability distribution with classical time-parameter Fisher information*

$$\boxed{I_{\mathrm{Fisher}}(t) = \sum_{j : q_j > 0} q_j \left(\partial_t \log q_j \right)^2 = 4 v_{\mathrm{FS}}(t)^2.} \tag{77}$$

The Hodge and Ebin–Marsden versions of a given pressure-balanced mode family obey $U_{EM}(t) = e^{4\nu t/R^2} U_H(t)$ *and therefore determine* identical *projective rays and identical* $q_j(t)$*. Their unnormalized energies and their Killing charges nevertheless differ.*

Proof. $\partial_t \log q_j = -2\nu(\lambda_j^2 - \overline{\lambda^2})$ gives the Fisher identity. The common scalar shift in each modal decay exponent cancels on projectivization. □

Corollary 12.3 (Complete logistic and geodesic reduction of the commuting Hopf–Killing pair)**.** *For* $k \ge 1$ *in Corollary 9.4, assume* $a, b \neq 0$ *and write the positive modal squared norms at* $t = 0$ *as* A_0, B_0*. Put*

$$\alpha = -2/R, \quad \beta = 2(k+1)/R, \quad \delta\gamma = \nu(\beta^2 - \alpha^2) = \frac{4\nu k(k+2)}{R^2} > 0. \tag{78}$$

The higher-frequency energy fraction $q(t) = B(t)/(A(t) + B(t))$ *and the projective mixing angle* $\theta(t) = \arctan\sqrt{B(t)/A(t)}$ *satisfy*

$$\boxed{\frac{q(t)}{1 - q(t)} = \frac{B_0}{A_0} e^{-2\delta\gamma t}, \quad \dot{q} = -2\delta\gamma q(1-q), \quad \tan\theta(t) = \sqrt{B_0/A_0}\, e^{-\delta\gamma t}.} \tag{79}$$

With $\mathscr{V} = M_2 - \mathcal{H}^2/W$ the signed-curl *helicity defect, the physically evolving ray obeys*

$$\boxed{v_{\mathrm{FS}}^2 = \nu^2(\alpha+\beta)^2 \frac{\mathscr{V}}{W} = \frac{4\nu^2 k^2}{R^2}\frac{\mathscr{V}}{W}.} \tag{80}$$

Its image is a meridional Fubini–Study geodesic, and its total length is exactly

$$\boxed{L_{\mathrm{FS}}([0,\infty)) = \arctan\sqrt{B_0/A_0}, \qquad \nu > 0.} \tag{81}$$

Thus the infinite-time projective path length depends on the initial mixture*, not on the numerical value of positive viscosity. At $\nu = 0$ the orbit is constant and the infinite-time limit does not commute with $\nu \downarrow 0$.*

Proof. The two modes are L^2 orthogonal since $\alpha \neq \beta$. Their real normalized coefficients, including their fixed initial signs, describe a single meridian in the projective line. Squared coefficients evolve at rates $2\nu(\alpha^2 - c_R)$ and $2\nu(\beta^2 - c_R)$, yielding (79). Since $\mathscr{V}/W = q(1-q)(\beta-\alpha)^2$ and $v_{\mathrm{FS}}^2 = \delta\gamma^2 q(1-q)$, the factorization $\beta^2 - \alpha^2 = (\beta-\alpha)(\beta+\alpha)$ gives (80). Integration of $|\dot\theta|$ gives (81). □

Proposition 12.4 (Energy–helicity inversion and a directly falsifiable time invariant)**.** *In the same two-mode sector, $W = 2\mathcal{E}$ and $\mathcal{H} = \alpha A + \beta B$ recover both instantaneous modal energies without velocity-field tomography:*

$$\boxed{A = \frac{\beta W - \mathcal{H}}{\beta - \alpha}, \quad B = \frac{\mathcal{H} - \alpha W}{\beta - \alpha}.} \tag{82}$$

For $A, B > 0$, the following observable is constant along the stipulated exact flow:

$$\boxed{\mathcal{I}_{\mathrm{rate}}(t) = e^{2\delta\gamma t}\,\frac{\mathcal{H}(t) - \alpha W(t)}{\beta W(t) - \mathcal{H}(t)} = \frac{B_0}{A_0}.} \tag{83}$$

The two subtractions in (82) are ill-conditioned when either energy fraction approaches zero: a relative error in a small modal energy can grow like the inverse of its fractional weight. Consequently, numerical reconstruction must reject or flag nearly extinct modes. Violation of (83), beyond the observational uncertainty and conditioning, rules out the exact two-mode ansatz with these fixed parameters. Conversely agreement does not *establish the ansatz uniquely amongst arbitrary fields.*

Proposition 12.5 (Finite-difference error amplification in energy-only projective tomography)**.** *Let $\ell(t) = \log W(t)$, and approximate $v_{\mathrm{FS}}^2(t) = \ell''(t)/4$ by $D_h = (\ell(t+h) - 2\ell(t) + \ell(t-h))/(4h^2)$. If measured values of ℓ have absolute errors at most ε and ℓ has four bounded derivatives on $[t-h, t+h]$, then*

$$\boxed{|D_h - v_{\mathrm{FS}}^2(t)| \le \frac{\varepsilon}{h^2} + \frac{h^2}{48}\sup_{|s-t|\le h}|\ell^{(4)}(s)|.} \tag{84}$$

Thus exact energy-only identifiability need not be numerically stable: taking $h \to 0$ at fixed noise increases uncertainty. This is a local deterministic estimate and makes no appeal to an unrelated inverse-boundary or deletion-trace theorem.

Proof. The centered second difference has fourth-order derivative remainder bounded by $h^2 \sup|\ell^{(4)}|/12$; division by 4 gives the bias. Three point errors contribute at most $(1+2+1)\varepsilon/(4h^2) = \varepsilon/h^2$. □

Nodal topology and interpretation. For $p_k = c(z_1 z_2)^k$ the *constituent* Hopf eigenfield w_k has nodal divisor consisting of the two linked Hopf fibres $z_1 = 0$ and $z_2 = 0$, each of multiplicity k; their Gauss linking number is $+1$ under the orientation used above. Decay does not move this divisor. It is *not* the nodal set of the full velocity U_k, since a nonzero Killing component generically persists on those fibres. Hence the exact chain here is a typed correspondence between a constituent nodal label, dissipative modal mixing, moment reconstruction and projective geometry, *not* a claim that fluid vortex lines braid spontaneously.

Prior art. The normalized heat/imaginary-time flow, spectral variance, Fubini–Study metric and Fisher information are standard constructions; see, e.g., work on imaginary-time Riemannian gradient descent and quantum geometry. The specific pressure-balanced S^3_R sector is the paper's specialization. This section asserts its derived identities, not first historical discovery of the underlying geometric-statistical relations.

13 Nonlinear leakage, projective speed and finite spectral moment certificates

We extend the energy–projective argument in Sec. 12 in two independent directions. The first identity is valid for arbitrary smooth *unforced* solutions, without pressure-balanced reduction. The second characterizes the additional positivity and finite-rank conditions satisfied by the exactly reducing histories. These are distinct logical statements: satisfying the scalar moment constraints is not sufficient to solve the nonlinear PDE.

13.1 Exact nonlinear Leray leakage from a projective–energetic residual

Theorem 13.1 (Nonlinear leakage identity for all smooth unforced solutions)**.** *Let (M, g) be a closed smooth Riemannian manifold, and A a fixed self-adjoint, nonnegative elliptic viscous operator on solenoidal fields. On any time interval of existence of a sufficiently smooth unforced Navier–Stokes solution with $W = \|u\|_2^2 > 0$, write*

$$\dot{u} = -\nu A u - N, \quad N = \mathbb{P}(\nabla_u u), \quad \langle N, u\rangle = 0. \tag{85}$$

Let $\psi = u/\sqrt{W}$, $v_{\mathrm{FS}} = \|\dot{\psi}\|_2$ and

$$\mu_A = \frac{\langle u, Au\rangle}{W}, \quad \sigma_A^2 = \frac{\|Au\|_2^2}{W} - \mu_A^2 \geq 0. \tag{86}$$

Then, pointwise in time,

$$\boxed{\frac{\|N\|_2^2}{W} = v_{\mathrm{FS}}^2 - \frac{1}{2}(\log W)'' + \nu^2 \sigma_A^2 \geq 0.} \tag{87}$$

An independent observable for the nonlinear viscous spectral-transfer contraction is

$$\boxed{\frac{1}{4}(\log W)'' - \nu^2 \sigma_A^2 = \frac{\nu}{W}\langle N, Au\rangle.} \tag{88}$$

If $\sigma_A > 0$ and $v_{\mathrm{visc}} = \nu \sigma_A$, this gives

$$\boxed{\|N\|_2 \geq \sqrt{W}\, \frac{|\frac{1}{4}(\log W)'' - \nu^2 \sigma_A^2|}{v_{\mathrm{visc}}}.} \tag{89}$$

Vanishing of the right-hand side of Eq. (88) does not *alone imply $N = 0$; by contrast Eq. (87) does. If $\sigma_A = 0$, the energy-curvature discrepancy vanishes even when the nonlinear projected field does not.*

Proof. For smooth solenoidal u on a closed manifold the energy cancellation gives $\langle u, N\rangle = 0$. Write $J = \langle u, Au\rangle$, $K = \|Au\|_2^2$ and $T = \langle N, Au\rangle$. Differentiation yields

$$W' = -2\nu J, \qquad J' = -2\nu K - 2T, \qquad W'' = 4\nu^2 K + 4\nu T. \tag{90}$$

Thus $(\log W)''/4 = \nu^2(K/W - J^2/W^2) + \nu T/W$, proving Eq. (88). Direct differentiation of ψ and energy cancellation give

$$\dot\psi = -\nu(A - \mu_A)\psi - \frac{N}{\sqrt{W}}, \quad v_{\mathrm{FS}}^2 = \nu^2\sigma_A^2 + \frac{2\nu T}{W} + \frac{\|N\|_2^2}{W}. \tag{91}$$

Eliminating T proves Eq. (87). Cauchy–Schwarz together with $\langle N, u\rangle = 0$ yields $|T| = |\langle N, (A-\mu_A)u\rangle| \le \|N\|_2\sqrt{W}\sigma_A$, proving Eq. (89). All differentiations use the assumed regularity. □

Remark 13.2 (Scope and experimental meaning)**.** *For Hodge viscosity on round S^3_R one may take $A = \Delta_H$; for the nonnegative Ebin–Marsden operator on the exact round sphere, $A = \Delta_H - 4R^{-2}I$. Equation (87) holds for either choice on any smooth existence interval. It does not derive a uniform bound for N from initial data and does not prove global regularity. The time-dependent projective state must be observed or computed; the energy alone cannot determine v_{FS} in arbitrary flows.*

13.2 A hierarchy of positive finite-rank moment certificates

Theorem 13.3 (Hankel spectral-rank hierarchy for reducing sectors)**.** *Let $u(t)$ be a pressure-balanced reducing history of Sec. 12, and aggregate all orthogonal modes with the same distinct positive decay* rate $\rho_1 < \cdots < \rho_r$, $\rho_j \ge 0$, *writing*

$$W(t) = \sum_{j=1}^{r} a_j e^{-2\nu\rho_j t}, \qquad a_j > 0. \tag{92}$$

For integers $n \ge 1$ define the continuous-time moment Hankel matrix

$$\mathsf{H}_n(t)_{ij} = \frac{(-1)^{i+j} W^{(i+j)}(t)}{(2\nu)^{i+j}}, \quad 0 \le i, j < n. \tag{93}$$

Then $\mathsf{H}_n(t)$ is positive semidefinite of rank $\min(n, r)$, *and*

$$\boxed{\det \mathsf{H}_r(t) = \Big(\prod_{j=1}^{r} a_j e^{-2\nu\rho_j t}\Big) \prod_{1\le i<j\le r} (\rho_i - \rho_j)^2 > 0.} \tag{94}$$

For any sampling interval $h > 0$, the discrete data $W_k = W(kh)$ and $q_j = e^{-2\nu\rho_j h} \in (0, 1]$ obey

$$\mathsf{G}_n = [W_{i+j}]_{i,j=0}^{n-1} \succeq 0, \qquad \operatorname{rank} \mathsf{G}_n = \min(n, r), \tag{95}$$

with $\mathsf{G}_n^+ = [W_{i+j+1}] \succeq 0$, $\mathsf{G}_n - \mathsf{G}_n^+ \succeq 0$ and the discrete complete-monotonicity inequalities $(-1)^d \Delta^d W_k \ge 0$. More precisely,

$$\boxed{\det \mathsf{G}_r = \Big(\prod_{j=1}^{r} a_j\Big) \prod_{i<j} (q_i - q_j)^2.} \tag{96}$$

The signs and ranks are necessary for Eq. (92), but not sufficient to certify the original fluid PDE from scalar measurements.

Proof. Let $V_{ij} = \rho_j^i$ and $D(t) = \mathrm{diag}(a_j e^{-2\nu\rho_j t})$. Then $\mathsf{H}_n = VD(t)V^\top$, and for $n = r$ the square Vandermonde determinant gives Eq. (94). For the sampled version replace ρ_j with q_j, proving the same factorization. The shifted and difference matrices have diagonal weights $a_j q_j$ and $a_j(1-q_j)$ respectively, and are therefore positive semidefinite. Finally $(-1)^d \Delta^d W_k = \sum_j a_j q_j^k (1-q_j)^d \geq 0$. □

Corollary 13.4 (Finite-energy-history Prony recovery and a chirality obstruction)**.** *In exact noiseless data, the first $2r$ values $W_0, \ldots, W_{2r-1}$ determine all q_j and their positive aggregate weights a_j uniquely, provided r is known. Indeed, for the monic polynomial $Q(z) = z^r + \sum_{\ell=0}^{r-1} c_\ell z^\ell$, solve*

$$\sum_{\ell=0}^{r-1} W_{i+\ell} c_\ell = -W_{i+r}, \quad 0 \leq i < r. \tag{97}$$

Its simple roots are q_j, and an ordinary Vandermonde solve gives a_j. The values $W_0, \ldots, W_{2r}$ additionally detect the exact finite rank by the first vanishing larger Hankel determinant. However, this procedure only identifies distinct squared-curl *rate shells, not curl chirality, eigenspace orientation, field phase or Hopf-link topology. Opposite-chirality eigenfields with equal $|\lambda|$ and equal norm have identical W_k for all times but opposite helicities.*

Proof. The annihilating polynomial obeys $\sum_{\ell=0}^{r} c_\ell W_{k+\ell} = \sum_j a_j q_j^k Q(q_j) = 0$ with $c_r = 1$. The coefficient matrix G_r is positive definite, hence invertible. Distinct nodes and positive weights imply uniqueness. Equal λ^2 have equal viscous rates for both operators under consideration and therefore cannot be separated by any energy-only time history, regardless of the number of samples. □

Theorem 13.5 (Viscosity-shift invariant normalized Hankel hierarchy)**.** *On the pressure-balanced round-sphere families, with the same initial field, the Hodge and Ebin–Marsden energies satisfy $W_{EM}(t) = e^{8\nu t/R^2} W_H(t)$. At equal sampling times kh, define dimensionless normalized data*

$$\boxed{R_k = \frac{W_k}{W_0}\left(\frac{W_0}{W_1}\right)^k, \qquad k \geq 0.} \tag{98}$$

Then each R_k and each Hankel minor of $[R_{i+j}]$ are exactly identical *for the two viscosity conventions. Their Hankel matrices remain positive semidefinite and have the same rank $\min(n, r)$. The condition $\mathsf{G}_n - \mathsf{G}_n^+ \succeq 0$ applies to the unnormalized nonnegative-decay data but need not apply to $[R_k]$, since normalization can move nodes above one.*

Proof. If $s = e^{8\nu h/R^2}$ then $W_k^{EM} = s^k W_k^H$ and $W_0^{EM} = W_0^H$. Substitution gives cancellation of every s^k in Eq. (98). Moreover $R_k = \sum_j (a_j/W_0)(q_j/\bar{q})^k$ where $\bar{q} = W_1/W_0 > 0$; the positive Vandermonde factorization applies to these rescaled positive nodes without requiring $q_j/\bar{q} \leq 1$. □

Proposition 13.6 (Rigorous sample-noise detection gate)**.** *Suppose each of the sampled energies $W_0, \ldots, W_{2n-2}$ has an absolute perturbation no greater than ε. Then the perturbed $n \times n$ Hankel matrix $\widetilde{\mathsf{G}}_n$ satisfies*

$$\boxed{\|\widetilde{\mathsf{G}}_n - \mathsf{G}_n\|_2 \leq n\varepsilon, \qquad |\sigma_j(\widetilde{\mathsf{G}}_n) - \sigma_j(\mathsf{G}_n)| \leq n\varepsilon.} \tag{99}$$

Consequently, if the smallest positive singular value of a rank-r ideal G_n exceeds $2n\varepsilon$, thresholding at $n\varepsilon$ recovers rank r. This is a sufficient *noise gate, not a uniform stability theorem as the spectral nodes collide or amplitudes vanish.*

Proof. Each of the n^2 entries of the perturbation matrix has absolute value at most ε, so its Frobenius norm is at most $n\varepsilon$, which bounds its spectral norm. Apply the singular-value perturbation inequality. □

Novelty and applicability gate. The Hankel/Prony methods are classical [1, 2]; the basic spectral-rank mechanism occurs in other heat-equation inverse problems, including a 2026 preprint [3]. The precise claims of this section concern their *typed application* to our exact round-S^3 Navier–Stokes reducing sectors, the dual viscosity normalization, the independent nonlinear Leray residual identity, and the explicit chirality no-go. This is not a claim that all these underlying ideas were discovered here. The moment inequalities remain necessary but not sufficient for PDE closure, and finite-dimensional moment identification is not a quantum Yang–Mills mass-gap argument.

14 Geometry of light as a typed, algebraically parameterized reconstruction system

This section extends the preceding nonlinear and phase-space certificates by a *joint but carefully nonidentified* optical–hydrodynamic reconstruction. It does not assert a new fundamental interaction. In particular, source-free Maxwell fields and viscous incompressible fluid flows are distinct physical equations, even if their initial data can be constructed from the same polynomial.

14.1 One polynomial, two equations, one nodal divisor

Let X_2, X_3 be a globally orthonormal pair in a positively oriented Hopf frame on the fixed round S^3_R, chosen consistently with the positive-curl convention. Let $p \in \mathrm{Sym}^m(\mathbb{C}^2)$ be nonzero and homogeneous of degree m, and suppose the null-Beltrami construction obeys

$$F_p = p(z_1, z_2)(X_2 + iX_3), \qquad \operatorname{curl} F_p = \lambda_m F_p, \quad \lambda_m = (m+2)/R, \quad F_p \cdot F_p = 0. \tag{100}$$

Here, the existence, orientation and global classification of such fields are part of the hypotheses inherited from the separately cited null-Beltrami construction; the present argument proves the consequences below, not that external classification afresh. Define $E = \operatorname{Re} F_p$, $B = \operatorname{Im} F_p$ and the normalized equal-time optical intensity

$$\mathcal{I}_p := \tfrac{1}{2}(|E|^2 + |B|^2) = |p|^2, \qquad |E| = |B|, \quad E \cdot B = 0. \tag{101}$$

The same E defines a distinct, *fluid* evolution $u(t) = e^{-\nu\lambda_m^2 t}E$ for Hodge viscosity, or $e^{-\nu(\lambda_m^2 - 4/R^2)t}E$ for Ebin–Marsden viscosity, because its convection is a gradient. On this exact family the fluid pressure, modulo an arbitrary function of time, is

$$\boxed{P(t,z) - P_0(t) = -\tfrac{1}{2}a(t)^2\mathcal{I}_p(z), \qquad u(t) = a(t)E.} \tag{102}$$

The optical field evolves by Maxwell's equations, *not* by the viscous factor $a(t)$. Equation (102) compares an equal-time intensity with a pressure profile from a different PDE; its precise common input is p.

Theorem 14.1 (Exact finite intensity/pressure tomography on the Hopf base)**.** *Write $p(z_1, z_2) = \sum_{k=0}^m c_k z_1^{m-k} z_2^k$. Assume its degree m and the nonzero intensity normalization are known. Choose any $m+1$ distinct latitudes $x_a \in (0,1)$ and $2m+1$ equidistant azimuthal angles $\delta_b = 2\pi b/(2m+1)$. The $(m+1)(2m+1)$ exact numbers $\mathcal{I}_p(x_a, \delta_b)$ determine the full Hermitian rank-one matrix $(c_k\bar{c}_l)_{k,l=0}^m$, and hence determine p up to one global $U(1)$ phase. They determine the associated divisor on $\mathbb{CP}^1$ and its Hopf-fibre lift, with multiplicities. The identical reconstruction applies to the pressure deficit $-2(P - P_0)/a(t)^2$ at fixed finite t, provided the pressure offset $P_0(t)$ and amplitude $a(t)$ are known (or independently calibrated).*

Proof. In Hopf coordinates $z_1 = \sqrt{1-x}\, e^{i\phi}$ and $z_2 = \sqrt{x}\, e^{i(\phi+\delta)}$, write

$$p = e^{im\phi} \sum_{k=0}^{m} c_k (1-x)^{(m-k)/2} x^{k/2} e^{ik\delta}. \tag{103}$$

Its intensity is a trigonometric polynomial in δ of degree at most m, so the specified $2m+1$ samples recover every Fourier coefficient without aliasing. For $d \geq 0$, its dth coefficient is

$$\widehat{\mathcal{I}}_d(x) = x^{d/2}(1-x)^{d/2} \sum_{l=0}^{m-d} c_{l+d}\overline{c}_l\, x^l (1-x)^{m-d-l}. \tag{104}$$

After division by the nonzero prefactor, the right side is a degree-$m-d$ polynomial. Evaluation at $m+1$ distinct latitudes determines its Bernstein coefficients uniquely. Varying d recovers the upper triangular entries $c_{l+d}\overline{c}_l$; Hermitian symmetry gives the others. Since $p \neq 0$, the matrix is positive rank one and factors as cc^*, uniquely modulo $c \mapsto e^{i\theta}c$. The fundamental theorem of algebra for homogeneous binary forms determines the unordered projective roots, counted with multiplicity. Their Hopf preimages are the nodal fibres. Equation (102) gives the pressure statement. No uniform numerical condition-number bound follows merely from invertibility. □

14.2 Temporal dual tomography and its information-theoretic blind spots

The preceding theorem reconstructs *spatial* structure. A separate temporal certificate reconstructs signed-curl population only on a finite pressure-balanced reducing family; it does not reconstruct general Maxwell fields or general turbulent fluids.

Theorem 14.2 (Finite joint energy–helicity Prony reconstruction)**.** *Let an exact reducing solution have a real orthogonal decomposition into r distinct* absolute *curl-eigenshells $\ell_j > 0$ with squared norms $A_{j,+}, A_{j,-} \geq 0$. Suppose the viscosity $\nu > 0$, radius R, time step $h > 0$, viscosity convention $c_R \in \{0, 4/R^2\}$, and the number of occupied shells r are known. Set $A_j = A_{j,+} + A_{j,-} > 0$ and $B_j = \ell_j(A_{j,+} - A_{j,-})$. From $2r$ exact samples of $W(kh) = 2\mathcal{E}(kh)$ and r exact samples of the helicity $\mathcal{H}(kh)$ one uniquely recovers, up to permutation, the nodes ℓ_j and both chiral weights $A_{j,\pm}$, provided the nodes are distinct. Specifically,*

$$\begin{aligned} W(kh) &= \sum_{j=1}^{r} A_j q_j^k, \quad \mathcal{H}(kh) = \sum_{j=1}^{r} B_j q_j^k, \\ q_j &= \exp[-2\nu(\ell_j^2 - c_R)h], \quad A_{j,\pm} = \tfrac{1}{2}(A_j \pm B_j/\ell_j). \end{aligned} \tag{105}$$

The reconstruction requires compatibility conditions $|B_j| \leq \ell_j A_j$; they are a falsification test for the claimed sector.

Proof. The positive Hankel matrix $[W((a+b)h)]_{a,b=0}^{r-1}$ is a Vandermonde–diagonal–Vandermonde product of rank r. The $2r$ moments determine the monic annihilating polynomial of its r distinct positive nodes by a nonsingular $r \times r$ Hankel solve. Its simple roots give the q_j; logarithms yield the ℓ_j^2 because ν, c_R, h are known and $\ell_j > 0$. Vandermonde inversion of the first r energy samples gives the A_j. The identical matrix inverted against the first r helicity samples gives B_j. Solving the two linear equations $A_j = A_{j,+} + A_{j,-}$ and $B_j = \ell_j(A_{j,+} - A_{j,-})$ gives the result. When nodes approach one another, the inverse is ill-conditioned; this theorem claims exact algebraic identifiability, not uniformly stable recovery. □

Proposition 14.3 (No-go for topology from integrated time traces alone)**.** *For every $m \geq 2$, the two binary forms*

$$p = z_1^m, \qquad q = \sqrt{m}\, z_1^{m-1} z_2 \tag{106}$$

have the same integral intensity on the unit round S^3, belong to the same signed curl shell under (100), and therefore generate identical normalized *energy, helicity and viscosity-only spectral time histories in the corresponding single-shell fluid construction. Nevertheless their projective root divisors are not equivalent: p has one root of multiplicity m, whereas q has two roots of multiplicities $(m-1,1)$. Hence integrated energy–helicity spectroscopy alone cannot reconstruct the Hopf nodal configuration, even with perfect noiseless data. Spatial intensity or pressure information is indispensable for this task.*

Proof. The Hopf-coordinate integral identity on round S^3 gives

$$\int_{S^3} |\sum_{k=0}^{m} c_k z_1^{m-k} z_2^k|^2 \mathrm{d}V = \frac{\mathrm{Vol}(S^3)}{m+1} \sum_{k=0}^{m} \frac{|c_k|^2}{\binom{m}{k}}. \tag{107}$$

For the displayed p and q the sum is one in both cases. They have the same m and therefore the same λ_m, while their projective roots have the different stated partitions. The modal energy and helicity traces then differ at most by the common convention-dependent overall normalization, which is equal for these forms. This exhibits the noninjectivity. □

Corollary 14.4 (Typed geometry-of-light reconstruction diagram)**.** *Within the hypotheses above there is a jointly compatible* typed *family of maps*

$$[p] \in \mathbb{CP}^m \longmapsto \mathcal{I}_p, \quad \mathrm{Div}(p), \tag{108}$$

$$\mathrm{Hopf}^{-1}(\mathrm{Div}(p)). \tag{109}$$

After fixing the signed-curl, viscosity, force and pressure conventions, a separate map yields the five quantities

$$u(t), \quad P(t), \quad W(t), \quad \mathcal{H}(t), \quad \mathcal{Z}(t).$$

The first map is injective through exact full spatial intensity; the map to integrated time traces is not. Every cross-sector equivalence requires an explicit, source-tracked geometric transformation and a test. *None of these statements asserts an equivalence between a viscous flow and a Maxwell spacetime solution, a detection of cosmic S^3, a general unforced Navier–Stokes theorem, or a quantum-gravitational relation.*

Priority and OpenAI Math transfer gates. Polynomial Maxwell knots and links predate this work, e.g. the exact solutions in de Klerk *et al.* and Bode; the Majorana/Husimi zero correspondence and optical stellar representations also predate it. The finite trigonometric/Bernstein and Hankel–Prony inversions are standard mathematical techniques. The present claim is a *joint certified application and a precise nonidentifiability obstruction* in the declared round-S^3 model; worldwide priority of that joint packaging has not been independently established. Family 350 of `openai/math` obstructs automatic transfer of round metric nodal conclusions to nearby nonround metrics. Family 376 concerns forced flat-domain Navier–Stokes computation and cannot close the unforced S^3 problem. Family 365 concerns boundary inverse data on a different geometric domain; its hypotheses do not apply to spatial intensity samples on a closed S^3. These three are useful *typed refusals*, not missing proofs that can be imported verbatim.

15 Catalogue transfer gates and relation to Millennium problems

Prior work on nonvanishing Beltrami eigenfields and the Jacobi/contact framework on the round sphere must also be credited [13]; thus the parity of curl eigenvalues alone is not asserted as a historically new discovery. For comparison, recent work has proved uniqueness/minimality results for the *static* Hopf map in a Faddeev–Skyrme regime [14]; this is distinct from the forced hydrodynamic construction here.

The public `openai/math` catalogue (accessed October 10, 2026) groups 719 manuscripts into 372 families; its supporting artifacts have differing verification scopes. Selected family numbers below are *research leads*, not imported lemmas without hypotheses.

Family	Relevant direction	Exact obstruction or required transfer
376	Universal computation in forced Navier–Stokes	Flat three-torus / flat-domain smooth forcing and prescribed control: no automatic S^3, unforced, or Clay regularity conclusion.
350	Nodal sets under near-round metrics	Counterexamples on metrics arbitrarily close to round S^3: protect exact-round assumptions; no uniform high-frequency nodal transfer.
130	Fast discrete Fourier transform	Exact complex arithmetic/unrestricted coefficients: no bit-complexity or curved-harmonic FFT bound without a new reduction.
365	Metric–connection inverse boundary data	Requires a genuine boundary and Dirichlet–Neumann map: a closed S^3 has no physical boundary from which that theorem follows.
290	Modular spectral recovery	Requires specified von Neumann factor, state and modular flow; coefficient-space Berry phase supplies none of them automatically.
362	Relativistic Vlasov–Maxwell	One-species nonnegative charge cannot be nontrivial on closed S^3 with smooth global Gauss law and no neutralizing background.
260	Spacetime Penrose rigidity	Asymptotic-flatness/end and trapping hypotheses not met by closed FLRW spatial S^3.

The core new conclusions of Secs. 6–11 come from explicit geometric identities and finite-dimensional Hilbert-space arguments; they do not assume that the catalogue manuscripts solve an unspecified transfer gap. A finite exact reducing family neither proves global regularity for arbitrary three-dimensional Navier–Stokes initial data nor settles the classical Yang–Mills quantum mass-gap problem. The positive spectral gap $2/R$ of classical curl on the round sphere is not a quantum Yang–Mills mass gap, and a Berry geometric phase does not prove the Hodge conjecture, $P \neq NP$, or the Riemann hypothesis. Formalization of *these* new hybrid theorems in Lean has not been performed.

16 A coupled but nonidentified research atlas

The two exact families have sharply different dynamical mechanisms. The viscosity-shift theorem relates their respective Hodge and Ebin–Marsden versions without identifying the two invariant families. In the unforced toroidal family, $\{a_n, b_n\}$ decay at unequal rates γ_n; the velocity is multi-eigenvalue and the nonlinear pressure contains $\gamma_n + \gamma_k$ beat rates. In the fixed-Hopf family, coefficients at the same m decay radially without forcing, and motion in $\mathbb{CP}^m$ requires external control. The shared invariant language is given by curl/Hodge spectral eigenvalues, exact pressure constraints, helicity/energy integrals, and finite-dimensional phase spaces. The former has a real linear Jacobi phase space and nonunique scalar pressure inversion; the latter admits rank-one pressure tomography and projective/Berry geometry. These should be visualized as separate nodes of a stratified atlas rather than a spurious universal isomorphism.

17 Validation and open mathematical questions

The accompanying V4 `verify_mathematics.py` checks 126 exact symbolic identities: for $n = 0, \ldots, 8$, the Jacobi curl-pair relations, Hodge eigenvalues, norms, and pairwise orthogonality. A three-mode numerical instance tests the reduced Navier–Stokes components, and the quadratic Hopf-braid example

verifies equality in Eq. (26) to machine precision. The V5 scripts `verify_commuting_hopf.py` and `verify_commuting_hopf_numeric.mjs` independently check all centralizer weights, signed curls, Lie brackets, explicit cross-product primitive and the complete inviscid nonlinear pressure balance for normalized quaternions; the symbolic tests reduce polynomial residuals modulo $\|q\|^2 - 1$. These are useful reproducibility tests, not independent formal proof certificates.

The curvature-viscosity bridge was checked independently for multi-rate modal solutions and cyclic projective forcing; see the companion verification script. Remaining research tasks include (i) nonlinear stability against non-invariant perturbations, (ii) pressure-only recovery with rate aliasing, (iii) actuator-limited optimization, and (iv) sharp lower, rather than upper, bounds on the cross-sector projected forcing. No statement here solves the unrestricted smoothness or blowup problem, certifies novelty against the entire literature, or proves that the universe is topologically S^3.

Declarations

This is a research working draft generated with AI assistance and requires independent expert review, author approval, attribution verification, and literature-priority evaluation before scholarly submission. No numerical or experimental measurements are claimed. A manuscript source file, exact symbolic tests, interactive research visualization, and a novelty audit accompany the draft.

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Table 1: The mathematical status of the proposed connections.

Sector	Exact derived structure	Limitation
Toroidal S^3	Jacobi multi-rate unforced solutions; pressure beat exponents; energy/helicity formulas	Not a classification of all flows; pressure inversion not unique
Commuting Hopf–Killing	Opposite-curl nonresonant exact flow; polynomial pressure; positive helicity defect with zero force	Only an explicit invariant centralizer, not all commuting eigenfields
Fixed Hopf	Prescribed polynomial path; pressure projectors; exact force cost	Phase control requires forcing
Cyclic braids	No-collision Hopf-link exchanges; Berry forcing bound	Braiding is not turbulence or general fluid-particle braiding
OpenAI Math 376	Smooth forced flat-domain computation mechanisms	No direct proof transfer or claim of shared theorem